THERMAL PROPERTIES OF MATTER

Important formulas and concepts directly from the NCERT –

❖ Temperature and heat-

- Temperature is a measure of "hotness" of a body.
- ***Heat is a form of energy*** transfer b/w two or more systems or a system and its surrounding due to "temp. difference".
- SI unit of temp = Kelvin(K)
  Commonly used unit of temp. = Celsius(^oC)

❖ Measurement of temperature-

- For ***mercury and alcohol***,volume varies linearly with temp. over a wide range.

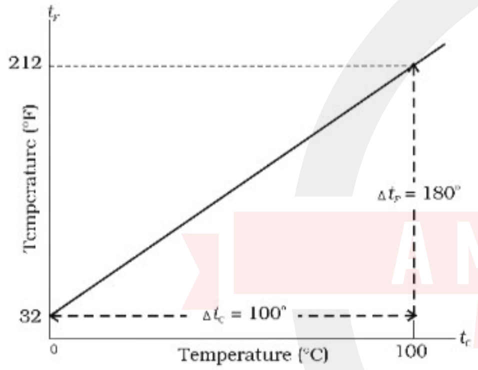

Fig. 10.1 A plot of Fahrenheit temperature (t_F) versus Celsius temperature (t_C).

- The equation of this graph is,

$$\frac{t_F - 32}{180} = \frac{t_C}{100}$$

- T= (t + 273.15) K if t is temperature in ^oC.
- Ideal gas eq^n ,

  $$PV = nRT \qquad (R = 8.31 \text{ J/mol K})$$

  It is applied to any quantity of any **low density gas**.
- Measurements on real gases deviate from the values predicted by ideal gas law at ***low temperature***.
- **Absolute zero** = -273.15 ^oC or "0 Kelvin".

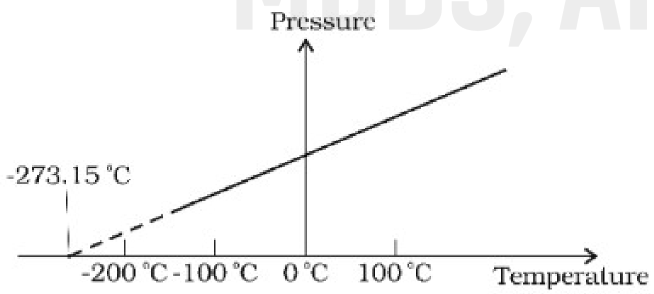

Fig. 10.2 Pressure versus temperature of a low density gas kept at constant volume.

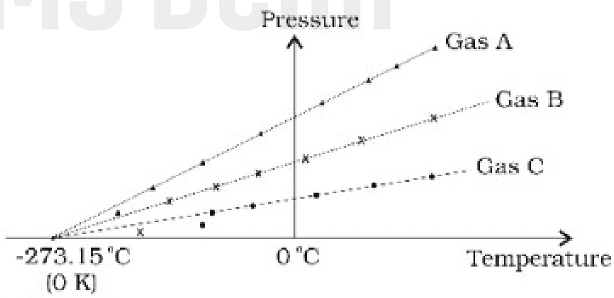

Fig. 10.3 A plot of pressure versus temperature and extrapolation of lines for low density gases indicates the same absolute zero temperature.

❖ **Thermal expansion**-

- Most substances expand on heating and contract on cooling.
- The increase in dimensions of a body due to the increase in its temperature is called thermal expansion.
- The fractional change in length ,$\Delta l/l$, is directly prportinal to ΔT.

$$\Delta l/l = \alpha_l \Delta T$$

α_l = *coefficient of linear expansion*

It is **characteristics of the material of the rod**.

- Copper expands five times more than glass for same rise in temperature.
- Normally, **metal expand more and have relatively high values of α_l.**
- Fractional change in volume,

$$\Delta V/V = \alpha_V \Delta T$$

α_V = *coefficient of volume expansion*

it is also **characteristics of substance** but **not strictly a constant,it depend in general on temperature.**

- **α_V becomes constant only at a high temperature.**

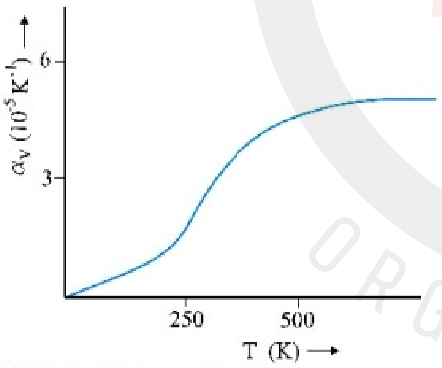

Fig. 10.6 Coefficient of volume expansion of copper as a function of temperature.

- Thermal expansion of solids and liquids is rather small,
  Materials like **pyrex and invar(a special iron nickel alloy) having low values of α_V.**
- **α_V** for alcohol(ethanol) is more than mercury and expands more than mercury for same rise in temp.
- **water exhibits an anomalous behavior , it has maximum density at 4°C.**
  Due to this property, lakes and ponds freeze at the top first.

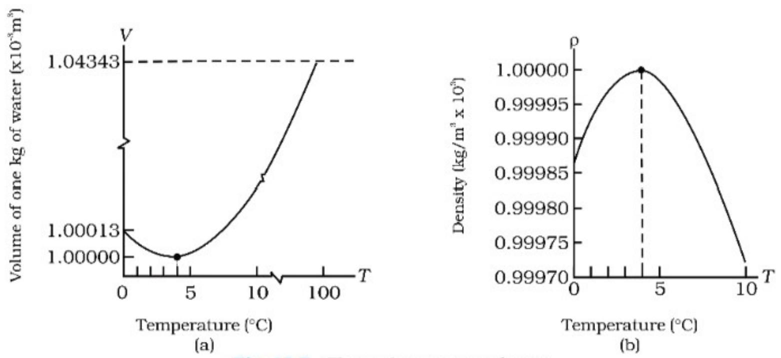

Fig. 10.7 Thermal expansion of water.

- **Gases at ordinary temp, expand more than solids and liquids(αV for gases are larger).**
- For liquids α_V is relatively independent of temp.
  For gases, α_V is dependent on temp. for an ideal gas,
  $\boldsymbol{\alpha_V = 1/T}$ **, it decreases with increasing temp.**
- Relation b/w coefficient of volume and linear expansion,
  $$\alpha_V = 3\alpha_l$$
- **Thermal stress = $Y_{steel}\,(\Delta l/l) = Y_{steel}\,\alpha_{l(steel)}\Delta T$**

❖ <u>Specific heat capacity</u>-

- Heat required to warm a given substance depends on **its mass , change in temperature and nature of substance.**
- **Heat capacity** = heat required to raise temp. of given mass by 1°C.
  $$H = \Delta Q / \Delta T$$
- **Specific heat capacity** –
  - It is the heat absorbed or given off to change temp. by unit of unit mass.
  - **$S = H/m = \Delta Q / m\Delta T$**
  - It is property of a substance**.**
  - **Depend on nature of substance and its temperature.**
  - SI unit is **J/kg K.**
  - Water has highest specific heat capacity ,due to which it is used as coolant in automobile radiators and heater in hot water bags.
    Due to this reason it heats and cools slowly.
- **Molar heat capacity –**
  - **Also depend on nature of substance and its temperature.**
  - SI unit = **J/mol K.**
  - **$C = \Delta Q/n\Delta T$**
  - At constant pressure, it is **C_P and** at constant volume, it is **C_V.**

❖ <u>Change of state</u> –

- **Melting point-**the temp at which solid and liquid states are in thermal equilibrium.
- **It is characteristic of a substance.**
- **It depend on pressure.**
- The melting point at **standard atm pressure** is called **'normal melting point'.**

- **Regelation -**due to pressure, wire passes through ice slab and slab doesn't split.this phenomenon of refreezing is known as 'regelation'.
  This happens due to fact that just below the wire, ice melts at lower temp due to increase in pressure**.**

- **Boiling point-**the temp at which liquid and vapour states of a substance coexist.
- **Boiling point increases with increase in pressure.**

- **Cooking is difficult on hills because,**
  At high altitudes,atmospheric pressure is lower,reducing the boiling point of water.
- **Cooking is faster in pressure cooker because,**
  Boiling point is increased inside a pressure cooker by increasing the pressure.

- **Sublimation-**the change from solid state to vapour state directly .
  Dry ice(solidCO_2) and iodine sublimes.

❖ **Latent heat** –

- The amount of heat per unit mass transferred during change of state of substance is 'latent heat'.
- Heat required during a change of state depends upon the heat of transformation and the mass of substance.

$$L = Q/m$$

  Unit of L = J/kg
- Value of L also depends on pressure.
- **Latent heat of fusion-**latent heat for solid liquid state change
  Latent heat of vapourisation-latent heat for liquid gas state change
- ***Heat is added or removed during a state change ,the 'temperature' remains constant.***

❖ **Conduction-**

- It is the mechanism of transfer of heat b/w two adjacent parts of a body because of their temperature difference.
- Gases are poor thermal conductors.
- Conduction may be described quantitatively as **rate of flow of heat or heat current(H).**

$$H = KA\,(T_2 - T_1)/L$$

  K = thermal conductivity

  Greater the value of K for a material,more rapidly will it conduct heat.

  SI unit – **J/s m K or W/m K**

- **Cooking pots have copper coating on bottom because**
  Being a good conductor of heat , copper promotes the distribution of heat over the bottom for uniform cooking.
- **Plastic foams are good insulators because they contain pockets of air.**
- **Concrete roofs are very hot during summer days due to its conductivity**
  Therefore, layer of earth or foam insulation on the ceiling is preffered so that heat transfer is prohibited.

❖ **Convection-**

- It is a mode of heat transfer by actual motion of matter.

- **It is possible only in fluids**.
- Convection can be natural and forced.
- In natural convection, **gravity** plays an important part.
- Convection involves bulk transport of different parts of the fluid.
- In forced convection,material is forced to move by a pump or by some other physical means.
- Common examples of forced convection are
  1)Forced air heating systems in home
  2)Human circulatory system
  3)Cooling system of an automobile engine.
- Examples of natural convection,
  1)Sea breeze
  2)Trade wind

❖ **Radiation** –
- It needs no medium
- **The energy so transferred by electromagnetic waves is called radiant energy**.
- In an electromagnetic wave(EM wave),electric and magnetic fields oscillate in space and time.
- **EM wave travels with speed of light(3×10^8 m/s)**
- **Electromagnetic radiation emitted by a body by virtue of its temperature,is called thermal radiation.**
- When this thermal radiation falls on other bodies, it is partly reflected and partly absorbed.
- **The amount of heat that a body can absorb by radiation depends on colour of the body.**
- Black bodies absorb and emit radiant energy better.
- **The bottoms of utensils are blackened so that they absorb maximum heat from fire and transfer it to the vegetables to be cooked.**

❖ **Blackbody radiation**-
- **The energy content for radiation varies for different wavelengths.**
- **The wavelength for which energy is maxiumum (λ_m) decreases with increasing temp.**
  This relation is given by wein's displacement law.

$$\lambda_m T = \text{constant}$$

- Value of wein's constant = **2.9×10^{-3} m K.**
- Blackbody radiation curves are **universal.**they depend only on temperature not on size,shape and material of blackbody.
- The total electromagnetic energy radiated by a body at absolute temp T is proportional to its size, its ability to radiate(emissivity) and temperature.

$$H = Ae\sigma T^4$$

  This is Stefan Boltzmann Law .

- **σ = Stefan Boltzmann constant**
  value = $5.67 \times 10^{-8} W/m^2K^4$
- **e = emissivity**
  e = 1 for a perfect radiator
  e = 0.4 for a tungsten lamp
- A body at temperature T with surroundings T_s , emits as well as receives energy. For perfect radiator, the net rate of loss of radiant energy is

$$H = \sigma A(T^4 - T_s^4)$$

For a body with emissivity e,relation

$$H = e\sigma A(T^4 - T_s^4)$$

❖ **Newton's Law for Cooling-**

- The law holds good for small difference of temperature.
- The loss of heat by radiation depends upon nature of the surface of the body and the area of the exposed surface.
- This law states that rate of cooling of a body is proportional to the excess temp over the surroundings,

$$dQ/dt = -k\ (T_2 - T_1)$$

T_1 = temp of surrounding
T_2 = temp of body

$$mSdT/dt = -k(T_2 - T_1)$$

$$dT/dt = -k(T_2 - T_1)/ms = -K\Delta T \quad (K = k/ms)$$

SUMMARY

1. Heat is a form of energy that flows between a body and its surrounding medium by virtue of temperature difference between them. The degree of hotness of the body is quantitatively represented by temperature.

2. A temperature-measuring device (thermometer) makes use of some measurable property (called thermometric property) that changes with temperature. Different thermometers lead to different temperature scales. To construct a temperature scale, two fixed points are chosen and assigned some arbitrary values of temperature. The two numbers fix the origin of the scale and the size of its unit.

3. The Celsius temperature (t_C) and the Farenheit temperare (t_F)are related by

$$t_F = (9/5)\ t_C + 32$$

4. The ideal gas equation connecting pressure (P), volume (V) and absolute temperature (T) is :

$$PV = \mu RT$$

where μ is the number of moles and R is the universal gas constant.

5. In the absolute temperature scale, the zero of the scale corresponds to the temperature where every substance in nature has the least possible molecular activity. The Kelvin absolute temperature scale (T) has the same unit size as the Celsius scale (T_c), but differs in the origin :

$$T_C = T - 273.15$$

6. The coefficient of linear expansion (α_l) and volume expansion (α_V) are defined by the relations :

$$\frac{\Delta l}{l} = \alpha_l \Delta T$$

$$\frac{\Delta V}{V} = \alpha_V \Delta T$$

where Δl and ΔV denote the change in length l and volume V for a change of temperature ΔT. The relation between them is :

$$\alpha_v = 3\,\alpha_l$$

7. The specific heat capacity of a substance is defined by

$$s = \frac{1}{m}\frac{\Delta Q}{\Delta T}$$

where m is the mass of the substance and ΔQ is the heat required to change its temperature by ΔT. The molar specific heat capacity of a substance is defined by

$$C = \frac{1}{\mu}\frac{\Delta Q}{\Delta T}$$

where μ is the number of moles of the substance.

8. The latent heat of fusion (L_f) is the heat per unit mass required to change a substance from solid into liquid at the same temperature and pressure. The latent heat of vaporisation (L_v) is the heat per unit mass required to change a substance from liquid to the vapour state without change in the temperature and pressure.

9. The three modes of heat transfer are conduction, convection and radiation.

10. In conduction, heat is transferred between neighbouring parts of a body through molecular collisions, without any flow of matter. For a bar of length L and uniform cross section A with its ends maintained at temperatures T_C and T_D, the rate of flow of heat H is :

$$H = K\,A\,\frac{T_C - T_D}{L}$$

where K is the thermal conductivity of the material of the bar.

11. Newton's Law of Cooling says that the rate of cooling of a body is proportional to the excess temperature of the body over the surroundings :

$$\frac{dQ}{dt} = -k\,(T_2 - T_1)$$

Where T_1 is the temperature of the surrounding medium and T_2 is the temperature of the body.

POINTS TO PONDER

1. The relation connecting Kelvin temperature (T) and the Celsius temperature t_c

$$T = t_c + 273.15$$

and the assignment $T = 273.16$ K for the triple point of water are exact relations (by choice). With this choice, the Celsius temperature of the melting point of water and boiling point of water (both at 1 atm pressure) are very close to, but not exactly equal to 0 °C and 100 °C respectively. In the original Celsius scale, these latter fixed points were exactly at 0 °C and 100 °C (by choice), but now the triple point of water is the preferred choice for the fixed point, because it has a unique temperature.

2. A liquid in equilibrium with vapour has the same pressure and temperature throughout the system; the two phases in equilibrium differ in their molar volume (i.e. density). This is true for a system with any number of phases in equilibrium.

3. Heat transfer always involves temperature difference between two systems or two parts of the same system. Any energy transfer that does not involve temperature difference in some way is not heat.

4. Convection involves flow of matter *within a fluid* due to unequal temperatures of its parts. A hot bar placed under a running tap loses heat by conduction between the surface of the bar and water and not by convection within water.